

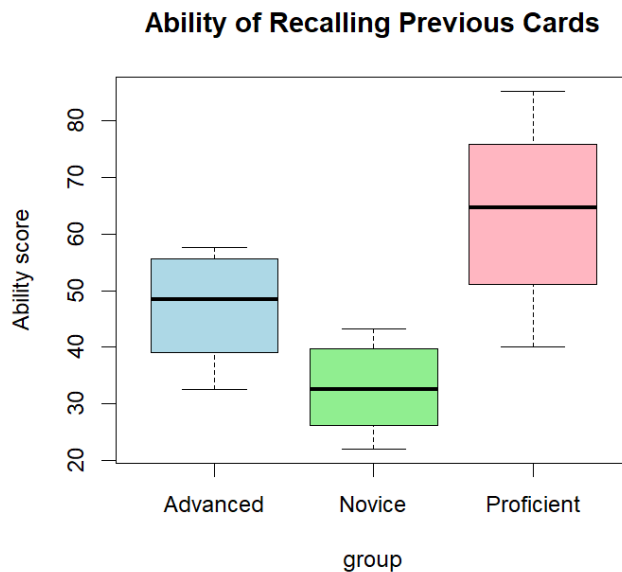
Msc Data Science
Probability and Statistics for Data Analysis
Assignment 2
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Exercise 1:

1.1.1 Descriptive Statistics

Group	n	mean	sd	median	IQR	min	Max
Novice	10	33.0	8.03	32.6	12.5	22.1	43.2
Advanced	10	46.8	9.03	48.4	15.5	32.5	57.7
Proficient	10	63.9	15.6	64.6	23.2	40.1	85.3

1.1.2 Visualization



The data were imported into the R environment and organized by players' experience level (novice, advanced, and proficient). Descriptive statistical analysis revealed a clear positive relationship between experience and recall ability: novice players exhibited the lowest mean scores, while proficient players demonstrated the highest. Notably, variability in performance increased with experience level, suggesting greater heterogeneity among more skilled players. Boxplot visualization confirmed these patterns, showing distinct differences in both median values and dispersion across groups. These findings support the hypothesis that experience level is significantly associated with recall ability.

1.2 Testing for differences among groups

To investigate whether recall ability differs among novice, advanced, and proficient players, the assumptions underlying parametric analysis were first examined. Normality within each group was assessed using the Shapiro-Wilk test, which indicated no significant deviations from normality for any group (novice: $p = 0.339$; advanced: $p = 0.319$; proficient: $p = 0.552$), satisfying this assumption.

Homogeneity of variances was evaluated using Levene's test, which revealed statistically significant differences across the three groups ($F(2, 27) = 5.90$, $p = 0.007$), indicating a violation of the equal variance assumption. Consequently, a parametric one-way ANOVA was deemed inappropriate.

Given the presence of heteroscedasticity, the non-parametric Kruskal-Wallis test was applied to examine differences in recall scores among groups. The analysis revealed a statistically significant effect of experience level on recall ability ($\chi^2(2) = 17.39$, $p < 0.001$), indicating that at least one experience group differs significantly from the others in recall performance.

1.3 Pairwise t-tests and validity of conclusions

To identify which specific groups differed, pairwise comparisons were conducted using Dunn's test with Bonferroni adjustment to control for multiple testing. The results revealed statistically significant differences between proficient and novice players ($p < 0.001$) and between proficient and advanced players ($p = 0.007$). The comparison between novice and advanced players did not reach statistical significance after adjustment ($p = 0.036$), exceeding the Bonferroni-corrected threshold ($\alpha \approx 0.0167$).

This approach is methodologically appropriate given the violation of homogeneity of variances detected in the preliminary analysis. Using standard parametric t-tests would be inappropriate in this context, as they assume equal variances across groups. Furthermore, conducting multiple uncorrected pairwise comparisons inflates the Type I error rate, increasing the likelihood of false-positive findings. The Bonferroni correction controls the family-wise error rate, ensuring that conclusions about group differences are robust and not artifacts of multiple testing.

In summary, the analysis followed a principled approach: a global non-parametric test (Kruskal-Wallis) to detect overall differences, followed by appropriately adjusted post-hoc comparisons (Dunn's test) to identify specific group differences while maintaining statistical rigor.

1.4 Identification of group differences

Following the significant Kruskal-Wallis test, post-hoc pairwise comparisons were conducted using Dunn's test with Bonferroni adjustment to control the family-wise error rate. The results indicated a statistically significant difference between novice and proficient players ($p = 9.30 \times 10^{-5}$). In contrast, no statistically significant differences were observed between novice and advanced players ($p = 0.076$) or between advanced and proficient players ($p = 0.161$) after adjustment. Therefore, at the 5% significance level, only the novice and proficient groups differ significantly in recall ability, while the advanced group does not differ significantly from either of the other two groups.

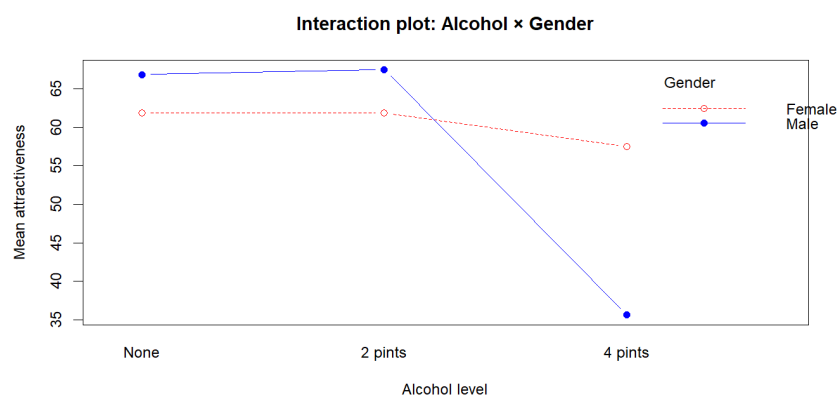
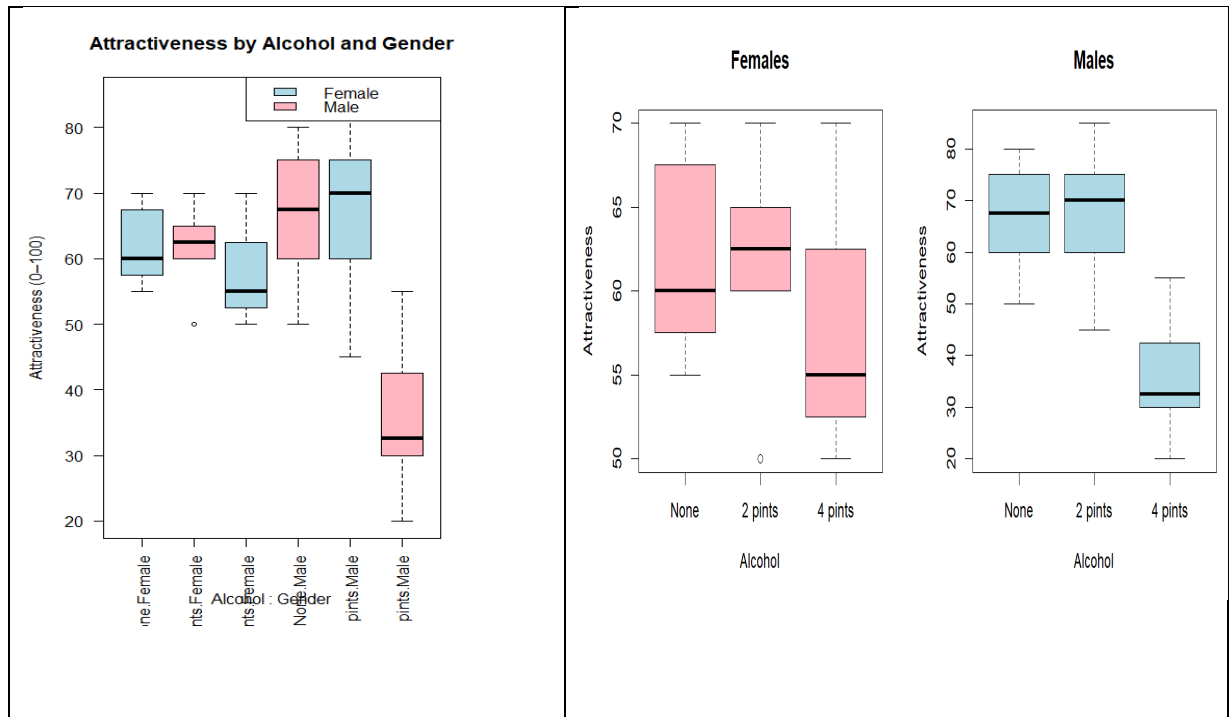
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Exercise 2:

2.1.1 Descriptive statistics

Alcohol	Female (n=8)	Male(n=8)
None	60.6 $\pm$ 4.96	66.8 $\pm$ 10.33
2 pints	62.5 $\pm$ 6.55	66.88 $\pm$ 12.52
4 pints	57.5 $\pm$ 7.07	35.63 $\pm$ 10.84

2.1.2 Visualization



Mean attractiveness scores for females remained relatively stable across all alcohol levels, showing minimal variation. In contrast, male attractiveness scores were similar at baseline (no alcohol) and moderate consumption (2 pints) but declined sharply at higher consumption (4 pints). Both boxplots and the interaction plot clearly illustrate this divergent pattern, revealing a significant gender-by-alcohol interaction: higher alcohol consumption substantially reduced perceived attractiveness for males, while females were largely unaffected.

2.2 Statistical model to describe the data

A two-way ANOVA with interaction:

$$Y_{ijk} = \mu + \alpha_i(\text{Alcohol}) + \beta_j(\text{Gender}) + (\alpha\beta)_{ij} + \varepsilon_{ijk}, \varepsilon_{ijk} \sim N(0, \sigma^2)$$

Where:

- α_i = main effect of Alcohol (3 levels)
- β_j = main effect of Gender (2 levels)
- $(\alpha\beta)_{ij}$ = interaction Alcohol x Gender

2.3 Test whether partner selection depends on alcohol and/or gender ($\alpha = 0.05$)

Two-way ANOVA

Source	DF	F	p-value
Alcohol	2	20.595	5.84e-07
Gender	1	2.018	0.163
Alcohol x Gender	2	11.789	8.64e-05

A two-way ANOVA was conducted to examine the effects of alcohol consumption and gender on partner attractiveness ratings. The assumptions for parametric analysis were first examined. The Shapiro-Wilk test applied to model residuals indicated no significant deviation from normality ($W = 0.982$, $p = 0.649$). Levene's test confirmed homogeneity of variances ($F(5, 42) = 0.94$, $p = 0.467$).

The ANOVA revealed a significant main effect of alcohol consumption ($F(2, 42) = 20.60$, $p < 0.001$), indicating that attractiveness ratings differed across alcohol levels. The main effect of gender was not statistically significant ($F(1, 42) = 2.02$, $p = 0.163$). Importantly, a significant interaction between alcohol and gender was observed ($F(2, 42) = 11.79$, $p < 0.001$), suggesting that the effect of alcohol on perceived attractiveness depends on gender.

2.4 Assessment of model assumptions

The assumptions of the two-way ANOVA model were assessed using formal tests and graphical diagnostics. Normality of residuals was evaluated using the Shapiro-Wilk test, histogram, and Q-Q plot. The Shapiro-Wilk test indicated no significant departure from normality ($W = 0.982$, $p = 0.649$), and graphical diagnostics showed residuals closely aligned with the theoretical normal distribution.

Homoscedasticity was examined using Levene's test across all combinations of alcohol level and gender. The test was not statistically significant ($F(5, 42) = 0.94$, $p = 0.467$), indicating that the equal variances assumption was satisfied. This was further supported by residuals versus fitted values and scale-location plots, which revealed no systematic patterns or variance changes.

Potential influential observations were investigated using Cook's distance. All observations fell below the commonly used threshold, suggesting no individual data point exerted undue influence on the fitted model. Overall, no serious violations of ANOVA assumptions were detected, and the model was deemed appropriate for inference.

2.5 Interpretation of the results in simple terms

The study examined whether alcohol consumption affects perceived attractiveness and whether this effect differs by gender. The results demonstrate that alcohol influences attractiveness in a gender-dependent manner. For women, attractiveness ratings remained relatively stable across all alcohol consumption levels. For men, attractiveness ratings were comparable at baseline and moderate consumption but declined substantially at higher consumption levels.

These findings indicate that higher alcohol consumption significantly reduces male attractiveness while leaving female attractiveness largely unaffected, revealing a clear interaction between alcohol and gender in perceived partner attractiveness.

Exercise 3:

3.1 Comparison of Reading Time Between Document Types

Document type	Number of document	Mean reading time(min)	Standard deviation(min)
General	372	12.56	1.68
Technical	228	13.02	3.41

This analysis examines whether technical documents take longer to read than general documents.

The dataset contains 600 documents, of which 372 are general and 228 are technical. The mean reading time for general documents is 12.56 minutes, while technical documents have a slightly higher mean reading time of 13.02 minutes. The difference in average reading time between the two groups is therefore about 0.46 minutes.

A two-sample t-test was used to compare the mean reading times of the two document types. The resulting p-value is 0.057, which is slightly above the 5% significance level. This means that the difference in mean reading time is not statistically significant.

In conclusion, although technical documents take slightly longer to read on average, there is not enough evidence to conclude that the mean reading time differs between general and technical documents.

3.2 Exploratory Analysis of Reading Time

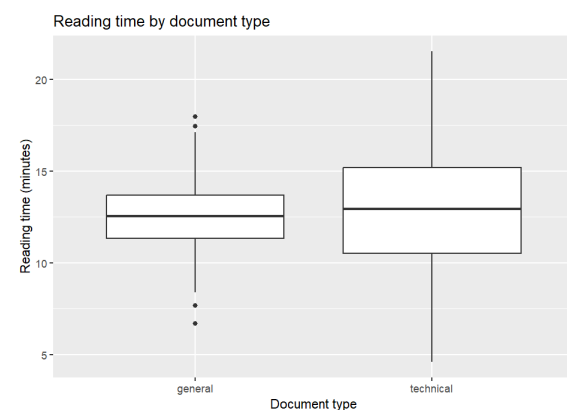


Figure 1 presents the distribution of reading times for general and technical documents using boxplots. The visualization reveals that technical documents exhibit moderately higher median reading times and greater variability compared to general documents. General documents display a more concentrated distribution, suggesting more consistent reading times across samples. Despite these differences, considerable overlap exists between the two distributions, indicating that reading time ranges are not entirely distinct between document types.

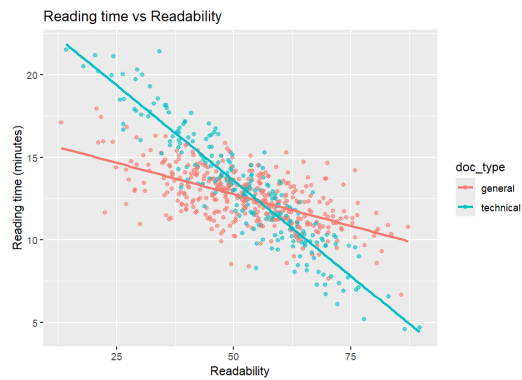


Figure 2 presents the relationship between reading time and readability, separately for the two document types. For both general and technical documents, higher readability scores are associated with lower reading times. This negative relationship appears stronger for technical documents.

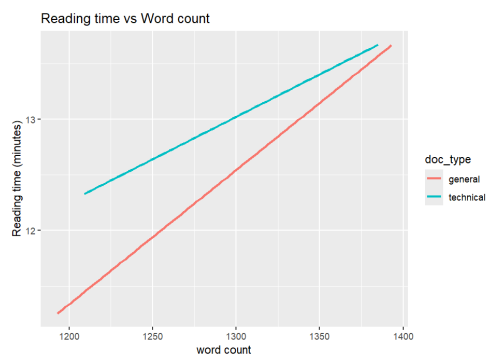


Figure 3 illustrates the relationship between word count and reading time across document types. Both general and technical documents exhibit a positive linear trend, with reading time increasing proportionally as word count rises. This pattern confirms the expected finding that longer documents require more time to read. Notably, the slopes of the trend lines are comparable between document types, suggesting that word count influences reading time similarly regardless of whether the content is general or technical in nature.

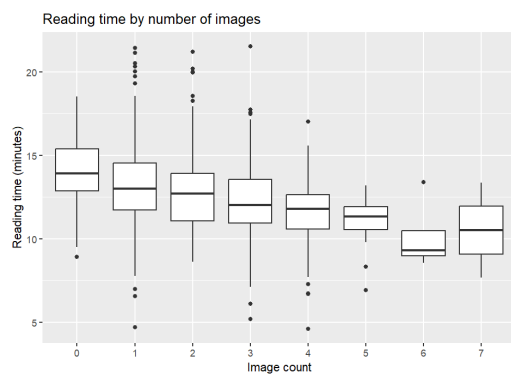


Figure 4 displays reading time by number of images. As the number of images increases, reading time generally decreases, although there is still considerable variability.

3.3 Fitted Linear Regression Model

A multiple linear regression model was fitted to explain reading time using word count, readability, number of images, and document type as explanatory variables.

Table reports the estimated regression coefficients together with their standard errors and p-values. All variables included in the model are statistically significant at the 5% level.

Variable	Estimate	Standard Error	p-value
Intercept	9.86	2.27	<0.001
Word count	0.0086	0.0017	<0.001
Readability	-0.139	0.0044	<0.001
Image count	-0.594	0.044	<0.001
Technical document	0.452	0.121	<0.001

The results show that word count has a strong positive effect on reading time, meaning that longer documents take more time to read. Readability has a negative effect, indicating that documents which are easier to read require less time. The number of images is also negatively related to reading time, suggesting that images may help readers process information more quickly.

The coefficient for document type is positive for technical documents. This means that, after controlling for word count, readability, and number of images, technical documents take on average about 0.45 minutes longer to read than general documents.

The overall fit of the model is strong. The R-squared value of 0.67 indicates that approximately 67% of the variability in reading time is explained by the model. This suggests that the selected variables capture a substantial portion of the factors influencing reading time.

Finally, using robust standard errors leads to similar conclusions, confirming that the results are statistically stable and not driven by outliers or heteroscedasticity.

3.4 Model Assumptions and Diagnostics

In this section, several diagnostic checks are used to assess whether the assumptions of the linear regression model are reasonably satisfied.

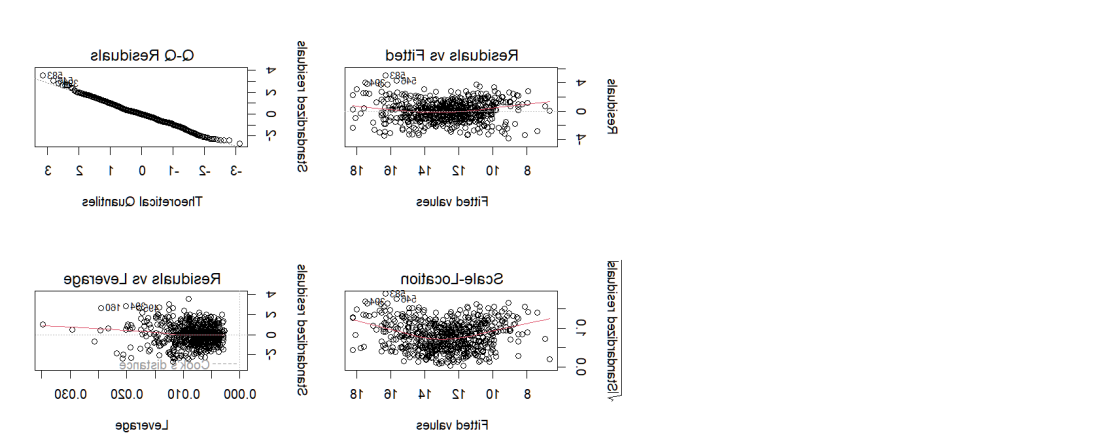


Figure 5 presents the standard regression diagnostic plots. The residuals versus fitted values plot does not show a strong systematic pattern, although there is some mild curvature and increasing spread for larger fitted values. This suggests that the linearity assumption is mostly reasonable, but there may be some heteroskedasticity.

The Q–Q plot of the residuals shows that the residuals follow the normal distribution fairly well, with only small deviations at the tails. This is supported by the Shapiro–Wilk test for normality, which gives a p-value of 0.124. Therefore, there is no strong evidence against the normality assumption.

The scale–location plot indicates that the variance of the residuals is not completely constant, as the spread of residuals slightly increases with the fitted values. This is confirmed by the Breusch–Pagan test, which results in a p-value of 0.001, suggesting the presence of heteroskedasticity.

Multicollinearity does not appear to be an issue in this model. All variance inflation factors are close to 1, indicating that the explanatory variables are not strongly correlated with each other.

Finally, Cook’s distance was examined to identify potentially influential observations. A small number of observations have higher Cook’s distance values, but none appear extreme enough to dominate the results. In addition, using robust standard errors leads to the same conclusions regarding statistical significance.

Overall, while there is some evidence of heteroskedasticity, the main assumptions of the linear regression model are reasonably satisfied, and the results of the model can be considered reliable.

3.5 Prediction of Reading Time

Regression Model Predictions: Reading Time Analysis

Using the fitted regression model, predictions were made for two documents with the same characteristics but different document types. Both documents have a word count of 1300, a readability score of 63.2, and 2 images. One document is general and the other is technical.

General Document

For the general document, the predicted reading time is approximately 11.1 minutes. The 90% prediction interval ranges from 8.7 to 13.5 minutes. This interval represents the range in which the reading time of a similar future document is expected to fall with 90% probability.

Technical Document

For the technical document, the predicted reading time is slightly higher, at approximately 11.5 minutes. The corresponding 90% prediction interval ranges from 9.2 to 13.9 minutes.

Interpretation

The predictions indicate that, for documents with the same characteristics, technical documents are expected to take slightly longer to read than general documents. However, the prediction intervals for the two document types overlap considerably, suggesting that the difference in reading time is relatively small.

Overall, the model provides reasonable predictions of reading time, although individual document reading times may still vary substantially.

Exercise 4:

Data and Objective

We use cross-sectional data from the HMDA dataset ($n = 2,380$), which contains information on mortgage applications. The dependent variable *deny* is binary and equals “yes” if the mortgage application was denied and “no” otherwise. Approximately 12% of applications in the sample are denied. The aim is to explain mortgage denial using applicant characteristics, financial ratios, credit history indicators, and local economic conditions.

Model Specification

Since the dependent variable is binary, a logistic regression (logit) model is employed. The logit model is appropriate because it models the probability of denial while ensuring predicted probabilities lie between zero and one. Coefficients are interpreted in terms of log-odds and odds ratios.

A full model including all available explanatory variables is first estimated.

Model Selection Using AIC and BIC

To identify the relevant explanatory variables, stepwise model selection based on the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) is applied. As expected, AIC selects a richer model, while BIC is more parsimonious due to a stronger penalty for model complexity.

The AIC-selected model retains key financial ratios, credit history variables, demographic characteristics, and local unemployment, suggesting these factors are most informative in explaining mortgage denial.

Effect of Marital Status (Single vs. Married)

According to the AIC-selected model, the estimated coefficient for *single* is positive and statistically significant. The corresponding odds ratio is: $\exp(\hat{\beta}_{single}) = 1.56$

This implies that, holding all other characteristics constant, single applicants have approximately 56% higher odds of mortgage denial compared to married applicants. This result reflects a statistically significant association between marital status and mortgage denial, conditional on observed applicant and loan characteristics.

Conclusion

Overall, mortgage denial is strongly associated with measures of credit history and financial risk. After controlling for these factors, marital status remains a significant predictor, with single applicants facing higher odds of denial than otherwise similar married applicants.

Exercise 5:

Data and Objective

This analysis examines the relationship between Covid-19 vaccination coverage and mortality across 27 EU countries using data from November 2021. For each country, we extracted the percentage of fully vaccinated adults and Covid-19 deaths per one million population over a 14-day period from a published figure. The objective is to model this relationship using multiple statistical approaches and identify the most appropriate method for these data.

Exploratory Analysis

Mortality varies substantially across countries, ranging from 0 to 325 deaths per million, while vaccination coverage ranges from 29% to 93%. A scatter plot reveals a strong negative relationship: countries with higher vaccination coverage consistently report fewer deaths. This pattern is confirmed by strong negative correlations (Pearson: -0.82 ; Spearman: -0.80).

Notably, increased variability at lower vaccination levels suggests heteroscedasticity, motivating the comparison of multiple modeling approaches to identify the most suitable framework for these data.

Linear Regression

We fitted a linear regression model as an interpretable baseline. Results indicate that each 1 percentage point increase in vaccination coverage is associated with approximately 4.8 fewer deaths per million over the 14-day period ($p < 0.001$). The model explains approximately 67% of the observed variability in mortality ($R^2 = 0.67$).

While straightforward to interpret, the linear model has important limitations: it does not account for the count nature of the outcome variable and may be sensitive to outliers, particularly at the extremes of the vaccination distribution.

Poisson and Negative Binomial Regression

Given that deaths are count data, we applied a Poisson regression model. The estimated rate ratio of approximately 0.95 suggests a 5% reduction in expected deaths per 1 percentage point increase in vaccination coverage. However, diagnostic tests revealed strong overdispersion (variance substantially exceeding the mean), violating a key assumption of the Poisson model and rendering standard errors unreliable.

To address overdispersion, we fitted a Negative Binomial regression model. This model estimates a rate ratio of approximately 0.93, corresponding to a 7% reduction in expected deaths per additional percentage point of vaccination coverage. This effect is statistically significant ($p < 0.001$) and the model appropriately accommodates the observed overdispersion, providing more reliable inference.

Model Comparison and Conclusion

Model comparison using the Akaike Information Criterion (AIC) strongly favors the Negative Binomial model, which provides the best fit to the data while appropriately handling overdispersion. The linear model remains valuable as a transparent baseline for communication purposes. Importantly, all three models consistently demonstrate a strong negative association between vaccination coverage and Covid-19 mortality, reinforcing the robustness of this finding across different statistical frameworks.

Limitations

Several important limitations should be noted. First, this analysis is observational and based on aggregated country-level data; therefore, causality cannot be inferred. Second, potential confounders such as population age structure, healthcare system capacity, testing policies, and non-pharmaceutical interventions are not included in the models. Third, the temporal relationship between vaccination timing and mortality outcomes is not explicitly modeled, though the 14-day mortality window provides some temporal separation. Finally, ecological analyses at the country level may not reflect individual-level relationships due to aggregation bias.